

Where shade reduces the thermal exposure of active mobility: an agent-based simulation study of twelve Spanish cities across five climate regimes

Federico Bellisardi^{1*} and José J. Ramasco¹

^{1*} Instituto de Física Interdisciplinar y Sistemas Complejos, IFISC (CSIC-UIB), CSIC – Universitat de les Illes Balears, Palma, Spain.

*Corresponding author(s). E-mail(s): fbellisardi@ifisc.uib-csic.es;

Abstract

DRAFT — results not yet available. Street-tree canopy and shade are among the few interventions that directly reduce the thermal exposure of people walking and cycling, but where to allocate a limited planting budget is an open, city-specific question, and the benefit of shade is not unconditional: the same canopy that cools a street in summer can add a small cold-stress penalty in winter. We study this joint heat-benefit/ cold-cost trade-off in twelve Spanish cities spanning five climate regimes (coastal Mediterranean, hot semi-arid, continental interior, highland, and Atlantic oceanic), using an agent-based mobility simulation (NOMAD) driven by real origin-destination demand (MITMA) on the real road network (OpenStreetMap), combined with real tree-canopy cover (ESA WorldCover, Urban Atlas), building height (Urban Atlas, CNIG/IGN PNOA LiDAR), a reproducible, data-driven calendar of representative days built from the full 2022 hourly thermal-comfort reanalysis (Copernicus UTCI, ERA5-Land), and in-situ meteorological observations (AEMET) for independent validation. We compare random, thermal-hotspot, flow-based, active-mobility-weighted, exposure-optimised and equity-weighted canopy allocation strategies at four intervention budgets (1%, 5%, 10% and 20% of street-edge length), evaluate each against both the cumulative heat dose it avoids and any cold-stress dose it adds, and report the resulting distribution of net benefit across a range of relative heat/cold weightings. [Abstract to be completed once results for all twelve cities are available.]

Keywords: urban heat, cold stress, thermal comfort, UTCI, active mobility, agent-based simulation, tree canopy, urban greening, equity, climate regimes, Spain

Urban heat exposure is not distributed evenly across a city’s street network, and neither is exposure to it: people walking or cycling are exposed to whatever thermal conditions exist along their route, for the whole duration of their trip, while people in cars are largely insulated from it. Tree canopy and shade are among the few interventions that directly reduce pedestrian- and cyclist-level thermal exposure at the street scale, but where to prioritise planting is a genuine allocation problem: canopy is costly to establish and maintain, and a given planting budget can be spent on the streets that are currently hottest, the streets that carry the most people, the streets that carry the most active (non-car) travel specifically, or some combination weighted by who lives nearby and how heat-vulnerable they are. A closely related remote-sensing study optimised urban greening allocation against population exposure to land-surface-temperature extremes across 200 cities worldwide ?; we address a complementary version of the same allocation problem using simulated agent-level active-mobility exposure (not residential population exposure from satellite-derived land surface temperature) as the objective, which lets us weight streets by who actually walks or cycles them and when, rather than by who lives nearby.

Two aspects of this allocation problem are, to our knowledge, rarely addressed together. First, shade is not an unconditional good: the same canopy that reduces UTCI on a hot afternoon also reduces it, by the same mechanism, on a cold winter morning, potentially trading a heat benefit for a cold-stress cost. Evaluating an intervention on a single hot reference day cannot reveal this trade-off. Second, Spain’s own climate diversity — from the Atlantic-oceanic north coast to the semi-arid south-east and the continental interior plateau — means that a single-city or single-region study cannot establish how general any given allocation strategy’s advantage is. We address both by (i) building a reproducible, data-driven calendar of representative days from the full year’s hourly UTCI series rather than picking dates by inspection, explicitly including cold-season and non-heatwave summer reference days alongside verified heatwave days, and (ii) applying the same pipeline, unmodified, to twelve Spanish cities chosen to span five climate regimes. We use real agent-based simulated mobility flows (not modelled trip generation alone) combined with real thermal-comfort and canopy data throughout.

1 Results

At the time of writing, the data and simulation pipeline described in Methods has been validated and run end-to-end for all twelve target cities. An initial car-only baseline on `QueueTrafficModel` was run for Palma de Mallorca (713,192 agents; 96.53% arrival rate, closely matching NOMAD’s own independently published validation figure of 96.5% for the same city and demand scale) and, after the threshold recalibration of Sec. 4.3, extended to all twelve cities. The main simulation campaign then moved to `LtmTrafficModel` at full demand scale across the reproducible calendar of representative days (Sec. 4.9), surfacing and fixing a genuine congestion-signal bug in that traffic model’s gate-wait accounting and adding pre-trip rerouting (Sec. 4.3); Table 2 reports the resulting teleport rate for every city, split by weekday- versus weekend/holiday-representative date, and the systematic gap between the two is discussed there and in Sec. 2.1.

Per-edge canopy fraction and per-edge, per-hour walk/bicycle flow (with exact conservation of routed demand, Sec. 4.6) have been computed for all twelve cities. The reproducible calendar-construction procedure (Sec. 4.9), applied to the real 2022 hourly UTCI series for all twelve cities, already yields two findings that inform the study design rather than its final conclusions: four of five candidate “ordinary summer” reference dates supplied for verification turn out to be locally in the 90th UTCI-max percentile or above (i.e. locally extreme) in at least one city despite falling outside every nationally-declared 2022 heatwave window, and conversely a substantial share (32.5%) of city-days inside a nationally-declared heatwave window are *not* locally extreme, concentrated in the two Atlantic-regime cities (Bilbao, A Coruña) — both consistent with the motivation in the introduction for verifying reference dates locally rather than only against a national declaration.

1.1 Canopy allocation strategy comparison

The six canopy-allocation strategies of Sec. 4.15 have been compared for ten of the twelve cities at the time of writing (all except Madrid and Barcelona: Madrid’s per-edge canopy/vulnerability/exposure tables were not yet built at the time of writing despite its car-traffic simulation being complete, Sec. 4.3; Barcelona’s canopy and pedestrian/cyclist flow data were completed most recently and its comparison was still running), each against a weekday-representative and a weekend/holiday-representative demand-and-thermal scenario built from the calendar of Sec. 4.9, at four canopy budgets (1%, 5%, 10%, 20% of street-edge length). Three consistent cross-city patterns emerge.

First, the two flow-aware, exposure-driven strategies (exposure-optimised and active-mobility-weighted) concentrate addressable heat/cold burden far more efficiently than the random baseline: at a 5% length budget, the random baseline addresses a share of a city’s total heat-stress burden essentially proportional to its length share (as it must, in expectation), while exposure-optimised and active-mobility-weighted address between 0% (in the three heat-negligible cases discussed below) and 86% of it with the same 5% of street length — a median concentration ratio of roughly 14 relative to random’s share among the seventeen combinations where the comparison is defined (i.e. excluding the three heat-negligible cases, where both random’s and the exposure-driven strategies’ share are zero) — across the 20 city/scenario combinations evaluated (ten cities times two representative-date groups). Thermal-hotspot allocation (ranking purely by peak UTCI, with no information about who actually walks or cycles there) is a considerably weaker proxy for this objective than either flow-aware strategy, addressing at most 32% of the same heat/cold burden at the same budget across every city and scenario tested — consistent with this study’s motivating premise that where it is hottest and where people actually experience that heat are not the same question.

Second, equity-weighted allocation trades a documented amount of raw exposure-efficiency for reaching more socio-demographically vulnerable populations: across the same 20 city/scenario combinations, its mean vulnerability index among selected edges exceeds the random baseline’s by 0.11–0.51 (median 0.25, on the index’s own [0,1] scale), i.e. routinely 50–100% higher vulnerability coverage than an unweighted

allocation, at a real but bounded cost in exposure addressed relative to the pure exposure-optimised strategy.

Third, and directly testing this study’s central heat/cold framing: for the three city/scenario combinations where ambient UTCI never crosses the moderate-heat threshold across the whole representative-date group (A Coruña, both scenarios; Bilbao, weekday) — an Atlantic-oceanic-regime finding consistent with the climate-regime grouping of Sec. 4.1 — the heat-only exposure term is exactly zero everywhere by construction, and exposure-optimised correctly and automatically falls back to minimising added cold-stress dose instead of producing an undefined or arbitrary ranking, selecting an allocation with dramatically lower cold-stress exceedance (by two to five orders of magnitude) than every other strategy in all three cases. Equity-weighted allocation, in the same three cases, illustrates a real and informative tension rather than a second instance of the same fallback: because it blends the (here, purely cold-driven) net-benefit term with the vulnerability index at equal weight by construction (Sec. 4.15), it is not guaranteed to also minimise cold cost, and in one of the three cases (Bilbao, weekday) it selects a materially *higher*-cold-cost allocation than even the random baseline because those higher-cold-cost edges happen to serve a more vulnerable population — a concrete, quantified instance of the cost side of an equity-weighted strategy that a heat-only objective could not have surfaced at all. Together, these confirm that the net heat/cold accounting of Sec. 4.13 is doing real work, not degenerating, exactly in the climate regime where a heat-only objective would have failed silently.

2 Discussion

[PENDING.]

2.1 Limitations

[DRAFT — to be expanded as the pipeline matures.] Shade is currently approximated as tree-canopy cover only; a genuine building-shadow model using the now-integrated CNIG/IGN PNOA and Urban Atlas building-height data (Sec. 4.12) is planned but not yet implemented, and in the interim a simplified evergreen/deciduous seasonal-canopy comparison (Sec. 4.14) uses illustrative, not species- or city-validated, leaf-on/leaf-off timing. The magnitude of canopy-shade UTCI reduction is a documented literature-range assumption, not a value calibrated against in-situ shaded/unshaded UTCI measurements for these specific cities. The heat/cold net-benefit trade-off (Sec. 4.13) depends on a relative-weight parameter λ that this study does not have grounds to fix on its own behalf; results are reported across a range of λ values rather than at a single chosen point. Barcelona’s road network was, for most of this study, blocked by a genuine infinite-loop defect in NOMAD’s own network-simplification preprocessing step (repeated re-contraction of already-absorbed nodes, triggered by Barcelona’s specific topology); this has since been diagnosed and fixed with an authorised, tested change to that preprocessing code (not to any simulation logic), and Barcelona’s network now builds in seconds rather than hanging, but its canopy, pedestrian/cyclist flow and intervention-strategy comparison were completed most recently among the twelve

cities and are reported in Sec. 1.1 only to the extent available at the time of writing. Madrid’s origin-destination matrix is roughly twenty times larger than Palma de Mallorca’s baseline, with a proportionally larger simulation cost; its car-traffic simulation is complete for the full calendar (Table 2), but its per-edge canopy, vulnerability and exposure tables needed for the intervention-strategy comparison were not yet built at the time of writing. NOMAD has no implemented public-transit model, so transit-mode MITMA trips are excluded from simulation entirely rather than reassigned to another mode. NOMAD’s native output instrumentation is car-only; pedestrian and cyclist flow relies on an offline-routing reconstruction (Sec. 4.6) that is exact under NOMAD’s own no-congestion-feedback design for those modes, but has not been cross-validated against a full discrete-event run with equivalent per-mode instrumentation, since NOMAD does not currently expose one. The socio-demographic vulnerability index currently has full census-section coverage only within each city’s core municipality, not its whole Functional Urban Area; edges outside that coverage fall back to the FUA-wide mean vulnerability rather than a locally estimated one. Eight of the twelve cities currently use the coarser, MITMA-zone-resolution, age-only vulnerability proxy rather than the INE income/poverty variant (Sec. 4.11), pending manual portal export of the remaining cities’ income/poverty tables. The “flow” strategy’s all-mode ranking (Sec. 4.15) is genuinely all-mode only where a car-traffic simulation has already been exported into the same per-edge table as pedestrian/cyclist flow; for cities where that join has not yet been performed, “flow” and “active-mobility” currently rank identically (both walk+cycle flow only), a documented, not silently absorbed, simplification for those cities. The systematic weekday-versus-weekend/ holiday teleport-rate gap of Table 2 — present in every city at a broadly similar order of magnitude despite very different city sizes and network topologies, which argues against a city-specific network or routing defect and toward a property of the weekend/holiday origin-destination matrix construction itself — is reported here as an open question rather than a resolved one; identifying its specific mechanism is ongoing work.

3 Conclusion

[PENDING — to be written once results are available for all twelve cities.]

4 Methods

Table 1 catalogues every real, third-party data source used in this study, its provider, its role in the pipeline, and its current integration status; it is kept as the single authoritative summary and updated whenever a new source is added. Each source is described in full, with its specific role and any documented simplifying assumptions, in the corresponding subsection below.

4.1 Study areas

Twelve cities — Palma de Mallorca, Zaragoza, Murcia, Valladolid, Madrid, Barcelona, Valencia, Sevilla, Córdoba, Granada, Bilbao and A Coruña — were selected to span both a range of city sizes and five Spanish climate regimes: coastal Mediterranean

Table 1 Data sources used in this study.

Source	Provider	Role	Status
OpenStreetMap	OSM contributors / Geofabrik	Road network topology	Integrated (11/12)
GHSL Functional Urban Areas	JRC / OECD	Study-area boundaries	Integrated (12/12)
MITMA mobility study	Ministerio de Transportes (Spain)	Origin-destination travel demand	Integrated (11/12)
ESA WorldCover 2021 v200	European Space Agency	Tree canopy cover	Integrated (11/12)
ERA5-Land	Copernicus C3S	Meteorological reanalysis	Integrated
Derived UTCI time series (full year)	Copernicus CDS	Ambient thermal comfort, calendar construction	Integrated (12/12)
AEMET OpenData	AEMET (Spain)	In-situ met. cross-check	Integrated (12/12)
Atlas de Renta de los Hogares	INE (Spain)	Vulnerability weighting	Integrated (municipality-level)
Urban Atlas (LC/LU, Street Tree Layer, Building Height)	Copernicus Land Monitoring Service	Land use, canopy cross-check, building height	Integrated (12/12)
PNOA LiDAR nDSM	CNIG / IGN (Spain)	Building/vegetation shadow	Integrated (12/12)

(Barcelona, Valencia, Palma de Mallorca), hot/semi-arid (Murcia, Sevilla), continental interior (Zaragoza, Valladolid, Madrid, Córdoba), highland (Granada), and Atlantic oceanic (Bilbao, A Coruña). This climate-regime grouping is a provisional working classification based on general geographic knowledge and is intended to be confirmed against a citable Köppen–Geiger dataset before being relied upon for any quantitative claim. Each city’s study area is its FUA, not its municipal boundary, following the Global Human Settlement Layer (GHSL) Functional Urban Areas delineation ?, sourced from the JRC/OECD GHSL data package (layer **FUAs**, matched by FUA name; see `network_prep/boundaries.py` in the code repository). All twelve cities now have a validated routing network, mobility demand and completed car-traffic simulation (Table 2); the per-edge canopy, vulnerability and pedestrian/cyclist-flow tables needed for the intervention-strategy comparison remain incomplete for Madrid and Barcelona at the time of writing (Sec. 2.1).

4.2 Simulation engine

Agent-based mobility simulation is performed by NOMAD (Network-based Open Mobility Agent Dynamics) ?, a discrete-event agent-based mobility simulator implemented in C++20, used here strictly as a black-box engine: its simulation logic (routing, traffic dynamics, event scheduling) is treated as fixed and is not modified for this study, except for one narrowly-scoped, tested addition to its Python bindings (exposing an existing C++ router/traffic-model attachment API and a deterministic per-edge, per-mode travel-time query, neither of which alters any simulation

behaviour) needed to extract per-mode flow that NOMAD’s own instrumentation does not otherwise expose (Sec. 4.6). NOMAD supports car, pedestrian and bicycle agents on a shared road-network graph derived from OpenStreetMap, with mode-aware routing (via A*) and road-class accessibility per mode, and a BPR volume-delay queueing traffic model for car congestion.

4.3 Stuck-agent handling and its calibration

An agent whose elapsed trip time exceeds a configurable multiple of its route’s free-flow time (`stuck_threshold_ratio`), or an absolute hard cap (`stuck_max_hours`), whichever binds first, is removed from the simulation (“teleported”) rather than left to run indefinitely; this mechanism is identical across NOMAD’s two available traffic models (`QueueTrafficModel`, a BPR volume-delay queue, and `LtmTrafficModel`, a Daganzo 1994 single-cell-per-link Cell Transmission Model with genuine receiving-side spillback), since both are invoked through the same `ITrafficModel` interface and neither implements this bookkeeping itself. An initial single-day, car-only pilot on `QueueTrafficModel` surfaced a real calibration issue in this project’s own wrapper (not in NOMAD’s simulation logic): the values this project’s code was passing (`stuck_threshold_ratio=20`, `stuck_max_hours=4`) had, in effect, only ever been exercised against Palma de Mallorca before this study’s twelve-city extension, and did not generalise — an otherwise identical car-only baseline for Zaragoza showed a 23.14% teleport rate at these values, against 1.28% for Palma. Relaxing the thresholds to a more moderate (50, 8h) pair, adopted as this project’s new default, reduced this to near zero for every city tested under that same single-day, off-peak-scale pilot.

That pilot, however, used `QueueTrafficModel` and a single reference day at reduced demand scale, and does not represent this study’s final results. The main simulation campaign uses `LtmTrafficModel` throughout, run at full demand scale across the reproducible calendar of representative days (Sec. 4.9), because it is the traffic model capable of genuine spillback (a downstream link at capacity blocks upstream discharge, rather than a car simply queueing invisibly past a full link as under the BPR volume-delay formulation) — the congestion signal the canopy-allocation objective of Sec. 4.13 needs to be realistic. Scaling `LtmTrafficModel` to full demand surfaced a second, distinct issue, this time in the traffic model’s congestion-signal bookkeeping rather than in this project’s threshold calibration: the model’s BPR-style travel-time term saturates once a link’s occupancy reaches its storage capacity (`on_enter` refuses further entries once full), so the reported travel-time ratio plateaus at a fixed ceiling ($\times 1.15$ in this implementation) no matter how many additional hours a fully-gridlocked link remains blocked — confirmed directly on a real Palma de Mallorca corridor edge that sat at $\sim 100\%$ storage occupancy for over sixteen continuous hours while its reported travel-time ratio never moved. A gate-wait counter already existed in the codebase to compensate for exactly this blind spot, but was wired only to an opt-in discharge-capacity mechanism, not to the spillback path that gates every link’s exit unconditionally; extending it to track continuous blocking from either cause (spillback or discharge-capacity exhaustion) fixed the signal without touching the underlying Cell Transmission Model dynamics themselves, and reduced Palma de Mallorca’s full-day

teleport rate from 39.5% to 24.1% in isolation. A complementary fix addresses a separate failure mode: every agent’s route is computed once, at simulation start, against free-flow costs, so an agent still waiting to depart hours later has no way to react to congestion that has since formed along its route; a bulk, parallel pre-trip rerouting pass, applied to each waiting agent shortly before its own scheduled departure using the same congestion-aware router already used for in-trip rerouting, closes this gap without the prohibitive cost of routing synchronously at every individual departure. With both fixes applied, the full twelve-city, five-date-calendar campaign (Table 2) shows a persisting, systematic weekday/weekend-or-holiday teleport-rate gap — median 1.16% on weekday dates versus 20.47% on the calendar’s two verified summer/holiday dates, roughly an order of magnitude higher, in every city without exception — which we report here as a genuine finding about demand-driven congestion under this study’s OD matrices rather than as a residual simulation artifact still to be eliminated; its root cause (most plausibly a property of the weekend/holiday OD matrix specifically, since the gap recurs at a similar order of magnitude across cities of very different size and topology) is not yet established and is noted as an open question in Sec. 2.1.

Table 2 Agent teleport rate under the final `LtmTrafficModel` configuration (spillback gate-wait fix, Sec. 4.3, plus pre-trip rerouting), full demand scale, weekday-representative vs. weekend/holiday-representative dates of the reproducible calendar (Sec. 4.9).

City	Teleport rate (weekday)	Teleport rate (weekend/holiday)
Palma de Mallorca	4.80%	25.21%
Zaragoza	0.42%	13.61%
Murcia	0.11%	3.45%
Valladolid	0.00%	0.02%
Madrid	6.37%	36.56%
Valencia	0.86%	23.87%
Sevilla	6.65%	33.02%
Córdoba	0.63%	10.00%
Granada	0.28%	8.41%
Bilbao	2.04%	20.47%
A Coruña	4.24%	25.04%
Barcelona	1.16%	17.16%
Median (all 12)	1.16% (weekday) / 20.47% (weekend/holiday)	

4.4 Road network

Road network topology for each city is extracted from OpenStreetMap [?], via regional `.osm.pbf` extracts published by Geofabrik GmbH, clipped to each city’s FUA polygon and simplified (degree-2 node contraction) by NOMAD’s own `simplify_osm.py` preprocessing script, called as a subprocess and not reimplemented (see `nomad.wrapper/build.py` in the code repository), with one authorised bug fix to that preprocessing step’s `NetworkCleaner` component (Sec. 2.1): it re-contracted

already-absorbed nodes indefinitely on Barcelona’s specific topology, an infinite loop rather than a slow computation, fixed by excluding nodes already absorbed by a prior contraction pass from being re-selected. For Palma de Mallorca this yields a routing graph of 79,111 nodes and 186,414 directed edges after simplification; graph sizes for the other eleven cities range from 26,471 nodes / 67,120 edges (Córdoba) to 495,697 nodes / 1,268,636 edges (Madrid); Barcelona’s now-buildable network is the second largest of the twelve at 917,528 edges.

4.5 Mobility demand

Origin-destination travel demand is derived from Spain’s Ministry of Transport (Ministerio de Transportes y Movilidad Sostenible, MITMA) mobile-phone-derived mobility study, publicly released as anonymised, aggregated daily zone-to-zone trip matrices [?]. Zone-level trips are converted to node-level, per-mode origin-destination pairs by NOMAD’s own `build_od.py` script (unmodified), which additionally restricts each mode’s candidate origin/destination nodes to that mode’s own largest strongly-connected component of the road network, so that every generated trip is guaranteed routable for its assigned mode, and which generates both an average-weekday and an average-weekend/holiday matrix (weekend/holiday classification via the same national-holiday calendar for Spain used throughout). Car mode trip counts are converted from MITMA person-trips to vehicle-trips using an average vehicle-occupancy factor of 1.20 persons per car (Spain average). For Palma de Mallorca, February 2022 average-weekday demand totals 1,922,543 person-trips across car (713,192), walk (573,925), bicycle (82,594) and public transit (552,832, excluded from simulation — NOMAD has no implemented transit model at time of writing).

4.6 Per-mode flow extraction

NOMAD’s own output instrumentation (its GeoJSON snapshot writer and in-memory link-state accessor) records occupancy, inflow and travel time only for car-mode agents: this is a property of the underlying traffic model, which is invoked only for car agents (pedestrians and cyclists never contend for car-calibrated link capacity, and NOMAD implements no pedestrian/bicycle congestion model), confirmed by direct inspection of the simulation engine’s source. Car-mode per-edge, 15-minute-bin flow, occupancy, mean travel time, free-flow travel time and derived congestion metrics are therefore extracted directly from NOMAD’s own periodic network-state snapshots. Pedestrian and bicycle per-edge, per-hour flow, person-seconds and travel time are instead obtained by direct routing (NOMAD’s A* router, mode-aware, queried through its Python bindings) rather than by running the discrete-event simulation: because neither mode ever experiences congestion feedback in NOMAD, each agent’s pre-computed route is deterministically its entire realised trip, so its per-edge timing can be computed exactly from the route and each mode’s free-flow speed, without the computational cost of discrete-event simulation. Per-agent departure times, sampled uniformly within each origin-destination row’s MITMA-derived hourly window, are propagated to per-edge, per-bin expected flow and dwell-time (person-seconds) contributions in closed form (exact overlap integrals of the departure window against each

time bin, not Monte Carlo sampling), so that the resulting per-edge-per-bin tables are exactly reproducible. Conservation of total routed demand (per-edge flow at trip origin equal to total routed agents; total person-seconds equal to the sum of in-simulation-window and next-day-spillover contributions) is checked programmatically for every extraction (see `nomad_wrapper/walk.bike.routes.py`), and has been confirmed exact for every city processed to date.

4.7 Tree canopy

Tree canopy cover is obtained from the European Space Agency’s WorldCover 2021 v200 global land-cover product [?](#), a 10 m-resolution global land-cover classification derived from Sentinel-1 and Sentinel-2 satellite imagery. Canopy fraction per road-network edge is computed as the fraction of “tree cover” (WorldCover class 10) pixels within a 15 m buffer of each edge’s centreline (straight line between its endpoint nodes, since the simplified routing graph does not retain intermediate street geometry), using zonal statistics against the relevant WorldCover tile(s) reprojected to each city’s local UTM zone (29N, 30N or 31N depending on longitude) for metric buffering. Mean edge-level canopy fraction varies substantially across the eleven cities processed to date, from 14.4–14.6% in the two semi-arid/ continental cities with the least canopy (Zaragoza, Murcia) to 25.5% in the Atlantic-regime city with the most (Bilbao), consistent with the climate-regime grouping of Sec. 4.1. The Copernicus Urban Atlas Street Tree Layer (Sec. 4.12) provides a second, independent per-city canopy source, now integrated but not yet cross-validated against the WorldCover-derived fraction used above.

4.8 Meteorological reanalysis and thermal comfort

Ambient (unshaded-baseline) thermal comfort is quantified using the Universal Thermal Climate Index (UTCI) [?](#), obtained directly as a derived reanalysis product from the Copernicus Climate Data Store’s `derived-utci-historical-timeseries` dataset [?](#), rather than recomputed from first principles, since the CDS product already applies the official UTCI methodology to ERA5 reanalysis inputs. The complete hourly UTCI series for calendar year 2022 (8,760 hourly values) has been retrieved for every one of the twelve cities in a single request each, rather than one request per reference day, after confirming directly against the live API that it accepts a genuine date-range request. Complementary raw meteorological fields (2 m temperature, 2 m dewpoint temperature, 10 m wind components, surface pressure, downward solar and thermal radiation) are obtained from the Copernicus `ERA5-Land` hourly reanalysis [?](#) for cross-validation and for driving a from-first-principles UTCI computation (via the `pythermalcomfort` implementation of the Bröde et al. [?](#) polynomial regression) where a real UTCI product is not available for a given city/date/scenario combination.

Street-level shade is approximated, for this version of the study, as tree canopy cover only (Sec. 4.7); building-shadow casting from the now-integrated building-height data (Sec. 4.12) is planned but not yet implemented (Sec. 2.1). The ambient UTCI is adjusted per road edge and hour by a canopy-shade correction, gated by actual solar elevation (computed from a standard closed-form solar-position equation for

each city’s latitude/longitude and simulation date, not from ERA5’s accumulated solar-radiation field, whose accumulation-reset convention at the 00 UTC time step introduces an ambiguity we chose not to rely on) so that the correction is zero at night and scales with daylight strength through the day. The magnitude of full-canopy-shade UTCI reduction at solar noon (default 6 °C) is an explicit, documented, calibratable assumption drawn from the range reported in the urban-microclimate literature for UTCI specifically (as opposed to mean-radiant-temperature reductions alone, which are typically reported as larger), not a value measured or fitted for these twelve cities.

4.9 Reproducible thermal calendar

Rather than selecting reference dates by inspection, we build a reproducible calendar of representative days directly from each city’s full 2022 hourly UTCI series. For each calendar month, we identify the day whose daily maximum UTCI is closest to that month’s median, either independently per city (“per-city medoid”) or, as our primary, recommended choice, as a single common date per month that minimises $\sum_c |\text{UTCI}_{\max}(c, d) - \text{median}_m(\text{UTCI}_{\max, c})| / \text{IQR}_m(c)$ across all twelve cities c for candidate day d in month m — directly comparable across cities since every city is evaluated on the same calendar date, at the cost of being individually less representative for any single city than that city’s own medoid; the per-city variant is retained as a robustness check. This yields twelve monthly reference dates for 2022. We additionally verify fifteen candidate dates supplied for the specific purpose of contrasting ordinary and extreme summer conditions — five “ordinary summer” dates and ten dates drawn from the two heatwaves officially declared by AEMET for summer 2022 (9–26 July and 30 July–15 August) — not by trusting the national declaration alone, but by computing each city-day’s UTCI-max percentile within that city’s own 2022 distribution, its meteorological season, and, for the minimum daily UTCI, an independent cold-stress category using the same official UTCI stress-category thresholds used throughout this study. The union of the twelve monthly dates and the fifteen verified dates, after removing one exact duplicate, gives a final 26-date calendar (Sec. 1); the minimum date subset needed for a main-text-scale analysis and the full 26-date set for a robustness appendix are both reported from the same underlying classification table.

4.10 In-situ meteorological observations

Daily station-level meteorological observations are obtained from AEMET (Agencia Estatal de Meteorología) OpenData [?](#), Spain’s national meteorological agency, for one representative station per city, selected as the station nearest each city’s true urban-centre landmark point (not the geometric centroid of its FUA polygon, which for irregularly-shaped FUAs can be materially offset from the city itself — confirmed for two cities in this study, where the nearest-by-centroid station would have been several kilometres further from the actual urban core than the nearest-by-true-centre station). These observations are used as an independent, real-world cross-check against the ERA5-Land/UTCI reanalysis products described above, not as a substitute input to the simulation pipeline.

4.11 Socio-demographic vulnerability

Heat-vulnerability weighting for the equity-oriented intervention strategy (Sec. 4.15) uses Spain’s National Statistics Institute (Instituto Nacional de Estadística, INE) Atlas de Distribución de Renta de los Hogares ?, a census-section-level household income and poverty-threshold indicator dataset, spatially joined to census-section boundaries retrieved live from INE’s own OGC API Features geometry service. Two real issues were found and corrected while integrating this source: the service’s response mixes true section-level polygons with one district-aggregate polygon per district that geometrically overlaps all of that district’s real sections, which is filtered out before the spatial join; and a flat single request silently truncates municipalities with more than 2,000 census sections (confirmed for Madrid, whose 2,483 real sections were being truncated to 2,000), fixed with standard OGC API Features pagination. At the time of writing, census-section coverage is available only for each city’s core municipality rather than its whole FUA (Sec. 2.1).

The Atlas de Renta’s income and poverty-threshold indicator tables themselves, unlike its census-section boundaries, are retrieved from INE’s table-query portal as a manual per-municipality export rather than a queryable API, and were obtained this way for only four of the twelve cities (Palma de Mallorca, Zaragoza, Murcia, Valladolid) at the time of writing. For the remaining eight, we substitute a coarser but fully automated proxy: each city’s population aged 65 and over as a share of total resident population, per MITMA “distrito” zone (the same official zonification and population reference already used for origin-destination demand, Sec. 4.5, joined to one day’s age-stratified “personas” file from the same source), spatially assigned to road-network edges by the same zone-to-edge join method as the income/poverty variant. Age-65-plus share is itself one of the three indicators the underlying composite-index formula is designed to accept (Sec. 4.15 uses it alone, at full weight, where income/poverty data is unavailable); MITMA’s zones are coarser than INE’s census sections, so this variant resolves vulnerability at zone rather than section granularity. This is a real, documented resolution and indicator-composition difference between the two groups of cities, not a value judgement about which population is more vulnerable, and is reported per city alongside the intervention results in Sec. 1.1.

4.12 Building height and detailed land cover

Two additional real data sources complement the canopy-only shade approximation of Sec. 4.7: the Copernicus Land Monitoring Service’s Urban Atlas ?, a high-resolution (≤ 0.25 ha minimum mapping unit) land-use/land-cover vector product for European Functional Urban Areas, together with its per-FUA Street Tree Layer and Building Block Height raster products, retrieved via the Copernicus Data Space Ecosystem’s S3 object store using a live per-product catalogue (rather than hardcoding file names, since each product’s version/date suffix is reprocessed over time); and Spain’s national mapping agency (Centro Nacional de Información Geográfica, CNIG/IGN) second-coverage PNOA airborne LiDAR-derived normalised digital surface models for buildings and vegetation ?, at 2.5m grid spacing, for which the exact set of national 1:50,000 map sheets intersecting each city’s FUA was computed from the official IGN

sheet grid rather than assumed from a fixed neighbourhood around each city’s centre — a real gap this correction closed: the fixed- neighbourhood approach previously in use under-covered even the four original study cities, most severely Palma de Mallorca, whose FUA extends across most of Mallorca island. A previously-documented limitation, that no usable building-height product could be found for these cities, is now known to have been a mis-scoped single national file (covering only the Canary Islands) rather than a genuine data-availability gap: the correct, per-FUA Urban Atlas Building Height product is available for all twelve cities. Both height products would allow a genuine building-shadow-casting shade model rather than the canopy-only approximation currently in use; this remains planned, not yet implemented (Sec. 2.1).

4.13 Thermal dose and shade benefit/cost accounting

For a given canopy scenario, city, and representative day, we define the cumulative heat and cold thermal dose experienced by active-mode travellers as

$$\text{heat_dose} = \sum_{e,t} N[e,t] \max(\text{UTCI}[e,t] - \theta_{\text{heat}}, 0) \Delta t, \quad \text{cold_dose} = \sum_{e,t} N[e,t] \max(\theta_{\text{cold}} - \text{UTCI}[e,t], 0) \Delta t,$$

where $N[e,t]$ is the walking-plus-cycling person-count on edge e during hour t (Sec. 4.6) and $\theta_{\text{heat}} = 26^\circ\text{C}$, $\theta_{\text{cold}} = 9^\circ\text{C}$ reuse the same official UTCI stress-category boundaries used throughout this study (onset of “moderate heat” and “slight cold” stress respectively), rather than newly chosen thresholds. Shade’s heat benefit and cold cost are then

$$\text{heat_benefit} = \text{heat_dose}_{\text{ambient}} - \text{heat_dose}_{\text{shaded}}, \quad \text{cold_cost} = \text{cold_dose}_{\text{shaded}} - \text{cold_dose}_{\text{ambient}},$$

and net benefit for a given relative weight λ is $\text{net_benefit}(\lambda) = \text{heat_benefit} - \lambda \text{cold_cost}$. Because λ encodes a value judgement this study has no independent grounds to fix, results are reported across a range of λ values (Sec. 2.1) and the two un-combined objectives (heat_benefit, cold_cost) are always reported alongside any λ -combined figure. This accounting has been validated end-to-end against real data for Palma de Mallorca: on 2022-02-08 (winter), an evergreen-canopy scenario shows zero heat benefit (as expected, ambient UTCI never approaches θ_{heat} in February) and a small positive cold cost, while a deciduous-canopy scenario (Sec. 4.14) shows zero cold cost on the same date, since a dormant deciduous canopy casts no shade to remove; on 2022-07-14 and 2022-07-20 (both inside the officially-declared July heatwave), both canopy types show an identical, substantial heat benefit and zero cold cost, as expected since a deciduous canopy is modelled as fully leafed in mid-summer.

4.14 Canopy phenology: evergreen versus deciduous

To assess whether the heat-benefit/cold-cost trade-off depends on planting evergreen versus deciduous species, we implement three canopy-type scenarios: a year-round (“evergreen”) shade fraction, a seasonally-attenuated (“deciduous”) shade fraction that ramps linearly to zero over a configurable window around leaf-off and back to full

shade around leaf-on, and an explicit “no distinction” scenario numerically identical to evergreen but naming the assumption. Real phenology literature exists for Mediterranean urban trees (e.g. studies of urbanisation effects on tree phenology in Madrid, and *Platanus* flowering-model studies for other Spanish cities), but extracting a single verified leaf-on/leaf-off day-of-year per city and species from that literature would require reading the full papers in depth rather than a title or abstract, which we have not yet done; consequently the current leaf-on/leaf-off timing (day-of-year 100 and 305, respectively, with a 20-day linear ramp) is an explicit, round-number placeholder for sensitivity analysis, not a value we claim is measured or validated for any of these twelve cities or their actually-planted species.

4.15 Intervention strategies

Six canopy-allocation ranking strategies are compared — random (null baseline), thermal-hotspot (highest peak UTCI first), flow (highest total all-mode flow first, where car-mode flow data is available; Sec. 1.1), active-mobility (highest pedestrian+cyclist flow first), exposure-optimised and equity-weighted — at four canopy budgets defined as a share of total street-edge length: 1%, 5%, 10% and 20%. Exposure-optimised ranks edges by net heat/cold benefit, $\text{exposure_score} - \lambda \text{cold_exposure_score}$, the per-edge analogue of Sec. 4.13’s $\text{heat_dose/cold_dose}$ applied to each edge’s actual (already-shaded, not ambient-vs-shaded) current thermal exposure rather than to shade’s marginal benefit — a deliberate choice, since an ambient-vs-shaded attribution is exactly zero by construction on any edge with little or no canopy today, precisely the edges a “where to plant next” ranking most needs to surface. At $\lambda = 1$ (the symmetric, equal-weight point already present in the λ -sweep of Sec. 4.13 and used here as the ranking default, not as a claim that this is the correct value), this reduces to exposure_score alone whenever $\text{cold_exposure_score}$ is negligible, and automatically pivots toward minimising $\text{cold_exposure_score}$ in the reverse case (Sec. 1.1). Equity-weighted allocation blends min-max-normalised net benefit and the vulnerability index of Sec. 4.11 at equal (0.5/0.5) weight. Each strategy/budget combination is evaluated once per city against a weekday-representative and a weekend/holiday-representative scenario, each an average over the respective dates of the reproducible calendar (Sec. 4.9) — a coarser two-scenario structure than the full per-date calendar, adopted for the results reported here for computational tractability across twelve cities; the full per-date and per-canopy-phenology-scenario (Sec. 4.14) evaluation remains future work (Sec. 2.1).

Abbreviations

AEMET: Agencia Estatal de Meteorología (Spanish State Meteorological Agency); CDS: Climate Data Store; CDSE: Copernicus Data Space Ecosystem; FUA: Functional Urban Area; GHSL: Global Human Settlement Layer; IGN: Instituto Geográfico Nacional; INE: Instituto Nacional de Estadística; MITMA: Ministerio de Transportes y Movilidad Sostenible; OSM: OpenStreetMap; PNOA: Plan Nacional de Ortofotografía Aérea; UTCI: Universal Thermal Climate Index.

Acknowledgements. This work uses NOMAD, an agent-based mobility simulator developed by F.B. ?. Mobility demand data were obtained from MITMA’s public mobility study; road network data from OpenStreetMap contributors under the Open Database Licence; land-cover and building-height data from ESA WorldCover and Copernicus Urban Atlas; meteorological reanalysis and thermal-comfort data from the Copernicus Climate Data Store; in-situ meteorological data from AEMET OpenData; socio-demographic indicators from INE; building/vegetation height from CNIG/IGN PNOA LiDAR.

Declarations

Funding. [AUTHOR INPUT REQUIRED: confirm funding sources, if any.]

Competing interests. The authors declare no competing interests.

Ethics approval. This study uses only anonymised, aggregated, publicly released third-party mobility, meteorological and land-cover data. No data collection from human participants was performed.

Availability of data and materials. This study uses the following third-party data sources, all publicly available:

- **Road network:** OpenStreetMap contributors, regional extracts via Geofabrik GmbH (<https://download.geofabrik.de>), Open Database Licence.
- **Functional Urban Area boundaries:** JRC/OECD Global Human Settlement Layer, Functional Urban Areas ?.
- **Mobility demand:** MITMA (Ministerio de Transportes y Movilidad Sostenible, Spain) mobility study (<https://movilidad-opendata.mitma.es>).
- **Tree canopy / land cover:** ESA WorldCover 2021 v200 ? (<https://esa-worldcover.org>), CC BY 4.0.
- **Meteorological reanalysis and UTCI:** Copernicus Climate Change Service, ERA5-Land ? and derived-utci-historical-timeseries ?, via the Copernicus Climate Data Store (<https://cds.climate.copernicus.eu>).
- **In-situ meteorological observations:** AEMET OpenData ? (<https://opendata.aemet.es>).
- **Socio-demographic vulnerability:** INE (Instituto Nacional de Estadística) Atlas de Distribución de Renta de los Hogares ? (<https://www.ine.es/dynt3/inebase/index.htm?padre=12385>), and census-section boundaries via INE’s OGC API Features geometry service.
- **Land use / street trees / building height:** Copernicus Land Monitoring Service Urban Atlas ? (<https://land.copernicus.eu/local/urban-atlas>), retrieved via the Copernicus Data Space Ecosystem.
- **Building/vegetation height:** CNIG/IGN PNOA second-coverage LiDAR-derived normalised digital surface models ? (<https://centrodedescargas.cnig.es>).

Aggregate results and processed data tables supporting this study will be made available upon publication.

Code availability. The simulation wrapper, data-processing pipeline and analysis code (`green_mobility`) will be made available in a public repository upon publication. The unmodified NOMAD simulation engine `?`, and the specific, narrowly-scoped, tested Python-binding addition used to extract per-mode flow (Sec. 4.6), are maintained in a separate repository.

Author contributions. [AUTHOR INPUT REQUIRED: confirm author contribution statement before submission.]